# Niccolò Cavallini

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## Education

**Swiss Federal Institute of Technology Zurich (ETHZ)** *Feb 2026 – Present*
*MSc in Computer Science — Major: Secure and Reliable Systems, Minor: Machine Learning*

- **Major Courses:** Automated Software Testing, Privacy Enhancing Technologies
- **Minor Courses:** Interactive Machine Learning: Visualization & Explainability, Philosophy of Language and Computation I
- **Other Courses:** Advanced Systems Lab, E-Business Law, Telecommunications Law
- **Currently Attending:** Information Security Lab, Network Security, Big Data

**Swiss Federal Institute of Technology Zurich (ETHZ)** *Sept 2021 – Jan 2026*
*BSc in Computer Science*

- **Bachelor's Thesis:** *Expert-in-the-Loop Visual Debugging for Structured Information Extraction from Free Trade Agreements* — **Grade: 5.75 / 6.00**
  Built an interactive visual debugging tool enabling domain experts to inspect and correct ML-based extraction of structured data (rules of origin, tariff concessions) from free trade agreement PDFs. Combined document parsing, information extraction, and a React-based UI for iterative expert feedback.
- **Electives:** Fundamentals of Web Engineering, Information Technology in Practice
- **Core Courses:** Computer Systems, Rigorous Software Engineering, Human Computer Interaction, Visual Computing, Introduction to Machine Learning

**Liceo Cantonale di Bellinzona** *Sept 2017 – June 2021*
*High school curriculum in Physics and Applications of Mathematics*

## Experience

**Owner & Full-Stack Developer** *Zurich*
*PCN Group di Niccolò Cavallini* *May 2024 – Present*

- Sole proprietor providing end-to-end IT services: requirements gathering, architecture, development, deployment, and ongoing maintenance for small businesses across Switzerland.
- **Fashion Boutique** — Designed and built a full-stack management platform in PHP: client registry, inventory and warehouse management, POS integration (receipt printing, cash drawer control), employee time tracking with detailed reports, loyalty program with gift cards and discount tiers, and customizable sales analytics dashboards.
- **Vacation Rental Property** — Developed a booking and client management system with integration into major booking platforms, automated email notifications for guests and owners, and Swiss QR-Invoice generation for billing. Included fiscal reporting and occupancy statistics.
- **Restaurant** — Built a responsive website with a CMS-backed menu system allowing the owner to update menus and a photo gallery independently.
- **Medical Practice** — Migrated sensitive health data (classified as *requiring special protection* under Swiss nFADP/nLPD), set up network infrastructure, connected medical equipment, and maintained a client-server practice management system.

## Projects

### Academic

**Expert-in-the-Loop Visual Debugging for Structured Information Extraction** *click here*

- Built an interactive visual debugging tool enabling domain experts to inspect and correct ML-based extraction of structured data (rules of origin, tariff concessions) from free trade agreement PDFs. Combined document parsing, information extraction, and a React-based UI for iterative expert feedback.
- Tools Used: Python, React, TypeScript

**Find Your Zurich Neighborhood** *click here*

- Developed a data-driven web app that recommends Zurich neighborhoods based on user lifestyle preferences (commute time, green spaces, nightlife, cost of living), aimed at helping newcomers navigate the city.
- Tools Used: TypeScript, PostgreSQL, Node.js

**Company (PCN Group)** — *click here*

**Retail Management Platform — Boutique Love My Style, St. Moritz** *click here*

- Built a complete retail management web application handling client records, inventory, point-of-sale with hardware integration (receipt printers, cash drawers), employee hour tracking, loyalty programs with gift cards, and sales analytics. Fully responsive and deployed in production.
- Tools Used: PHP, JavaScript, Node.js, MySQL

**Booking & Management System — Cà Val Forno (Vacation Rental)**

- Developed a booking and client management system with integration into major booking platforms, automated email notifications for guests and owners, Swiss QR-Invoice generation, and fiscal reporting with occupancy statistics.
- Tools Used: PHP, JavaScript, MySQL

**Website — Restaurant Alpenrose, Sils**

- Built a responsive website with a CMS-backed menu and photo gallery, allowing the owner to update content independently.
- Tools Used: PHP, JavaScript

**Infrastructure & Data Migration — Studio Medico Loviglio, Locarno**

- Migrated sensitive health data (classified as *requiring special protection* under Swiss nFADP/nLPD), set up network infrastructure, connected medical equipment, and maintained a client-server practice management system.

## Technologies

**Languages:** Java, PHP, Python, JavaScript/TypeScript, SQL, Visual Basic .NET, C (basic), C++ (basic)

**Frameworks & Libraries:** React, Node.js, FastAPI, WordPress

**Databases:** PostgreSQL, MySQL, Redis

**Tools & Infrastructure:** Git, Docker, Linux, Microsoft Intune, Networking (LAN/WLAN setup)

## Languages

**Italian:** Native **English:** Fluent **German:** Intermediate **French:** Basic